Module 1: Cryptocurrency and Regulations

1. Discuss cryptography with basic terminology.
2. What is Cryptocurrency? Explain the working of cryptocurrency.
3. A software company needs to send confidential files from its headquarters to a remote branch. Both locations use the same encryption key to encrypt and decrypt files both before and after transmission. Explain the working of the symmetric cipher model to ensure the security of the files during transmission.
4. What is Hash? Explain the use of a hashing algorithm in blockchain.
5. A user accesses an online banking application. The session between the user's device and the bank's server is encrypted using a symmetric cipher. The session key is agreed upon at the beginning of the session. Describe how symmetric encryption is used during the session to protect user data.
6. Explain Asymmetric Key Cryptography with a suitable example.
7. Explain Symmetric Key Cryptography with a suitable example.
8. A healthcare network uses blockchain to store patient records. The doctor digitally signs each transaction, updating a patient’s record before being added to the blockchain. The system ensures that only authorized doctors can make changes. Interpret how the digital signature verifies the authenticity of the doctor’s transaction with a suitable diagram.
9. An election system is implemented on a blockchain where each voter’s choice is digitally signed to guarantee that only their valid vote is counted, and that it cannot be tampered with afterward. Explain how digital signatures ensure that each vote is traceable and non-repudiable.
10. Summarize the History of Cryptocurrency. List names of cryptocurrencies used in blockchain application development.

Module 2: Introduction to Blockchain

1. Demonstrate the Core blockchain architecture components with a suitable diagram.
2. Compare and contrast public blockchains and private blockchains with suitable examples.
3. Define various features of blockchain. Also, explain the role of the distributed ledger in the blockchain network.
4. Discuss smart contracts and explain the working of smart contracts for below scenario mentioned. **(Any one scenario)**

   Scenario 1: A real estate platform uses a smart contract to transfer property ownership once both buyer and seller digitally approve the transaction and payment is verified.

   Scenario 2: A freelancing platform implements smart contracts where funds are released to freelancers only after the client confirms project completion based on agreed terms. Illustrate how smart contracts ensure that payments are tied to specific deliverables.
5. Compare Centralized vs. Distributed Ledger
6. Analyze the role of a consensus algorithm in a blockchain network. Discuss the importance of consensus in maintaining the integrity and security of the blockchain.
7. Explain the term Immutability in a blockchain network with a suitable example.
8. UAI is planning to start an Electric Vehicle (EV) charging station in the parking area and intends to implement a verifiable billing system using blockchain technology. Apply the blockchain and implement a block diagram with challenges, key components, and user roles.
9. Explain Proof of Work (PoW) and Proof of Stake (PoS)- working, example, pros/cons.
10. Explain Blockchain Technology Stack with a suitable diagram.

Module 3: Bitcoin and Decentralization

1. What is Bitcoin? How Bitcoin Works
2. Bitcoin Script & Execution
3. Bitcoin Transaction and Collection
4. Transaction Signature and Parameters

5. Transaction Validation Rules
6. Bitcoin Mining and Block Structure
7. Bitcoin mining incentives
8. What is Ethereum? How Ethereum works: Ethereum network is a peer to peer public network, which allows any individualor organization to take part or create account in it.
9. Explain features of Ethereum
10. Explain DApp architecture using Blockchain and P2P technologies. Decentralized Vs. Distributed Applications
11. Trust and Security in DApps

Module 4- Distributed Consensus

1. Byzantine General Problem and Byzantine Fault Tolerance
2. Explain: Difficulty and Target is used to measure how difficult to mine a block in bitcoin network
3. What is Smart Contract? A simple model for smart contracts With an increasing trend towards autonomous applications, smart contracts are fast becoming the preferred mechanism to implement financial instruments.
4. How do smart contracts work? (Realestate)
5. What is Consensus Algorithms in blockchain. List out the names of Consensus Algorithms.
6. What is Proof of Work (PoW)? Explain working of PoW algorithm step by step.
7. What is Proof of Stake (PoS)? Explain working of PoW algorithm step by step.
8. What is Delegate Proof of Stake (PoS)? Compare PoS and Delegate PoS.
9. What is Proof of Burn (PoB)? Explain working of PoB algorithm step by step.
10. What is Proof of Capacity (PoC)? Explain working of PoC algorithm step by step.

Module 5- Blockchain Use Cases

Consider below Use cases for module 5 questions.

- Land Registration
- Critical Infrastructure
- Electric Vehicle Charging Infrastructure
- Aadhar Access

1. Why is Blockchain Technology our future of **(any use case)**?
2. How does Blockchain Technology works in **((any use case)**?
3. How Distributed Ledger Technology (DLT) is important in **((any use case)**?
4. Benefits of Blockchain technology that may prove to be useful for businesses in **(any use case)** industries.
5. Blockchain Applications That Are Transforming Society in **(any use case)**
6. Business Financial Services in **(any use case)**
7. Areas of **(any use case)** in which blockchain can improves its functionality